

Command-line networking reference

Three commands that tell you if it's **reachable**, **where it breaks**, and **who you are** on the network.

ping, tracert, and ipconfig are the first tools any technician reaches for when a connection misbehaves. Each answers a different question – together they cover most of what you need to diagnose a network problem from a single terminal.

cmd.exe

```
C:\> ipconfig
IPv4 Address. . . . . : 192.168.1.42
C:\> ping 8.8.8.8
Reply from 8.8.8.8: bytes=32 time=14ms TTL=117
C:\> tracert google.com
Trace complete. 5 hops to 142.250.x.14
```

ping Packet InterNet Groper

Tests whether a host is reachable and measures round-trip time. Sends ICMP Echo Requests to a target and waits for Echo Replies – the network equivalent of shouting into a hallway and timing the echo.

Syntax

```
ping [options] destination
```

Examples

```
ping google.com By hostname
```

```
ping -t 192.168.1.1 Continuous, until  
stopped
```

```
ping -n 10 8.8.8.8 Send exactly 10  
requests
```

```
Pinging google.com [142.250.x.14] with 32 bytes of data:
```

```
Reply from 142.250.x.14: bytes=32 time=14ms TTL=117
```

```
Reply from 142.250.x.14: bytes=32 time=13ms TTL=117
```

```
Reply from 142.250.x.14: bytes=32 time=15ms TTL=117
```

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)
```

Result	Meaning
Replies, low ms	Host is reachable, connection is healthy
Request timed out	No reply – host down, blocking ICMP, or unreachable
High / inconsistent ms	Network latency or congestion
100% packet loss	No working path between source and destination

- Confirm a device, server, or website is online
- Test the local link to your router/gateway
- Diagnose latency and packet loss
- Verify DNS resolution (pinging a name resolves it to an IP first)

tracert traceroute on Linux/macOS

Maps the path your data takes to a destination, listing every router (hop) along the way and the time to reach each one. It works by sending packets with increasing TTL values – each router that "expires" a packet reveals itself, one hop at a time.

Syntax

```
tracert [options] destination
traceroute [options] destination
```

Examples

```
tracert google.com Windows
-----
traceroute google.com Linux / macOS
-----
tracert -d google.com Skip hostname
                        lookups
```

```
Tracing route to google.com [142.250.x.14]
over a maximum of 30 hops:

  1      2 ms      1 ms      1 ms  192.168.1.1
  2     10 ms      9 ms     10 ms  10.10.0.1
  3     15 ms     14 ms     15 ms  isp-gateway.net
  4     22 ms     21 ms     22 ms  core-router.isp.net
  5     14 ms     13 ms     14 ms  142.250.x.14

Trace complete.
```

Result	Meaning
Steady, low times	Healthy route, hop by hop
Spike at one hop	Congestion or slowdown at that point
* * * (timeout)	That hop didn't reply – may still forward traffic fine
Trace stops early	A firewall or routing issue partway through the path

- Pinpoint exactly where a connection is failing – your network, ISP, or the destination
- Diagnose slow connections by isolating which hop adds delay
- Understand the actual route traffic takes across the internet

ipconfig

ifconfig / ip addr on Linux/macOS

Shows your own device's network configuration – IP address, subnet mask, gateway, and DNS servers. Usually the first command run in any troubleshooting session, since it tells you what your device thinks its own setup is.

Syntax

```
ipconfig [options]
ifconfig
ip addr show
```

Examples

```
ipconfig /all  Full details incl. MAC,
                DNS, DHCP
-----
ipconfig /release  Release current DHCP
                    lease
-----
ipconfig /renew    Request a new IP address
-----
ipconfig /flushdns  Clear local DNS cache
```

Ethernet adapter Ethernet:

```
IPv4 Address. . . . . : 192.168.1.42
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
```

Field	Meaning
IPv4 Address	Your device's address on the local network
Subnet Mask	Defines the size/boundary of the local network
Default Gateway	The router your device sends outbound traffic through
DNS Servers	Used to resolve domain names into IP addresses

- Find your device's own IP address and network settings
- Diagnose IP conflicts or misconfiguration
- Renew or release a DHCP lease when a connection misbehaves
- Flush stale DNS entries that point to the wrong address

Which tool answers which question

ping

"Is it up, and how fast?"

Checks reachability and latency to one specific

tracert

"Where exactly is the problem?"

ipconfig

"What are my own settings?"

target.

Maps every hop between you and the target to isolate the failure point.

Reveals your device's IP, gateway, and DNS configuration.

A typical troubleshooting order

1 Check your own configuration

Run `ipconfig` – confirm you have a valid IP, gateway, and DNS.

2 Ping the gateway

Confirm you can reach your local router.

3 Ping an external IP

e.g. `ping 8.8.8.8` – confirms internet connectivity, bypassing DNS.

4 Ping a domain name

e.g. `ping google.com` – confirms DNS resolution is also working.

5 Trace the route

If a step above fails, run `tracert` to find exactly where the path breaks.